INSUFFICIENT AIR (OR NO AIR) BLOWN FROM VENTS [FULL-AUTO AIR CONDITIONER]

id0703c1700300

DESCRIPTION	• Problem with each vent and/or duct • Airflow mode does not change
POSSIBLE CAUSE	• INTERIOR 15 A fuse malfunction • Open or short circuit in wiring harness between battery positive terminal and climate control unit • Open or short circuit in wiring harness between climate control unit and airflow mode actuator / air mix actuator • Airflow mode actuator malfunction • Airflow mode link malfunction • Airflow mode door malfunction • Vent or duct malfunction (clogging, deformation, air leakage, improper operation)

- When performing an asterisked (*) troubleshooting inspection, shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunctions. If there is a problem, check to make sure connectors, terminals and wiring harnesses are connected correctly and undamaged.

Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT FUSE** • Remove the INTERIOR 15 A fuse. • Is the fuse normal?	Yes	Go to the next step.
		No	**If the fuse is melted:** Repair or replace wiring harness for short to ground and replace the fuse. **If the fuse is deterioration:** Replace the fuse. Then go to Step 19.
2*	**INSPECT TO SEE WHETHER MALFUNCTION (OPEN CIRCUIT) IS IN B+ SIGNAL WIRING HARNESS (BETWEEN FUSE BLOCK AND CLIMATE CONTROL UNIT) OR ELSEWHERE** • Disconnect the climate control unit connector. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Switch the ignition ON (engine off or on). • Measure voltage at the climate control unit terminal L. • Is the voltage **approx. 12 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and fuse block terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the Step 19.
3*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN A/C UNIT OR AIRFLOW MODE ACTUATOR** • Switch the ignition off. • Connect the climate control unit connector. • Switch the ignition ON (engine off or on). • Measure voltage at the climate control unit terminal I when airflow mode at VENT and DEFROSTER. • Is voltage normal?	Yes	Go to Step 15.
		No	Go to the next step.
4*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN POSITION SENSOR OR ELSEWHERE** • Measure voltage at the climate control unit terminal G when temperature setting to MAX HOT and MAX COLD. • Is voltage normal?	Yes	Go to Step 7.
		No	Go to the next step.

Step	Inspection		Action
5*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY OR SHORT TO POWER SUPPLY OR GROUND) IS IN POSITION SENSOR POWER SUPPLY OR ELSEWHERE** • Measure voltage at the climate control unit terminal H. • Is voltage **approx. 5 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and airflow mode actuator / air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 19.
6*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY OR OPEN CIRCUIT) IS IN POSITION SENSOR GROUND OR ELSEWHERE** • Measure voltage at the climate control unit terminal E. • Is voltage **below 1.0 V**?	Yes	Go to Step 10.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and airflow mode actuator / air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 19.
7*	**INSPECT TO SEE WHETHER MALFUNCTION (SHORT TO POWER SUPPLY) IS IN WIRING HARNESS (AIRFLOW MODE POSITION SIGNAL) OR ELSEWHERE** • Measure voltage at the climate control unit terminal I. • Is voltage **approx. 12 V**, at Step 3?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and airflow mode actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to the Step 19.
		No	Go to the next step.
8*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN WIRING HARNESS (AIRFLOW MODE POSITION SIGNAL) OR ELSEWHERE** • Switch the ignition off. • Disconnect the climate control unit connector. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Disconnect the airflow mode actuator connector. (See AIRFLOW MODE ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Verify continuity at between climate control unit terminal I and airflow mode actuator terminal D. • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and airflow mode actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 19.

Step	Inspection		Action
9*	**INSPECT TO SEE WHETHER MALFUNCTION (SHORT TO GROUND) IS IN WIRING HARNESS (AIRFLOW MODE POSITION SIGNAL) OR POSITION SENSOR COMMON (POWER SUPPLY OR GROUND) HARNESS** • Verify continuity at between climate control unit terminal I and ground. • Is there continuity?	Yes	Repair wiring harness between climate control unit terminal and airflow mode actuator terminal. Then go to Step 19.
		No	Inspection and repair for open circuit following: • Between airflow mode actuator terminal B and junction point to each air mix actuator (position sensor power supply). • Between airflow mode actuator terminal A and junction point to each air mix actuator (position sensor ground). Then go to Step 19.
10*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN AIRFLOW MODE ACTUATOR WIRING HARNESS (BETWEEN CLIMATE CONTROL UNIT AND AIRFLOW MODE ACTUATOR) OR ELSEWHERE** • Measure voltage at the following terminals of climate control unit. — Terminal V (DEFROSTER mode motor drive signal) — Terminal T (VENT mode motor drive signal) • Is voltage normal?	Yes	Go to the next step.
		No	Go to Step 12.
11*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN AIRFLOW MODE ACTUATOR OR WIRING HARNESS (BETWEEN CLIMATE CONTROL UNIT AND AIRFLOW MODE ACTUATOR)** • Switch the ignition off. • Disconnect the climate control unit and airflow mode actuator connectors. • Verify continuity at following terminals between airflow mode actuator and climate control unit. — Climate control unit terminal V — airflow mode actuator terminal B (DEFROSTER mode motor drive signal) — Climate control unit terminal T — airflow mode actuator terminal A (VENT mode motor drive signal) • Is there continuity?	Yes	Go to Step 15.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and airflow mode actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 19.
12*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN AIRFLOW MODE ACTUATOR OR ELSEWHERE** • Switch the ignition off. • Disconnect the airflow mode actuator connector. • Switch the ignition ON (engine off or on). • Measure voltage at the following terminals of airflow mode actuator harness-side connector. — Terminal B (DEFROSTER mode motor drive signal) — Terminal A (VENT mode motor drive signal) • Is voltage normal?	Yes	Go to Step 15.
		No	Go to the next step.

Step	Inspection		Action
13*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (SHORT TO B+ BETWEEN CLIMATE CONTROL UNIT AND AIRFLOW MODE ACTUATOR) OR ELSEWHERE** • Switch the ignition off. • Disconnect the climate control unit connector. • Switch the ignition ON (engine off or on). • Measure voltage at the following terminals of the climate control unit harness-side connector. — Terminal V (DEFROSTER mode motor drive signal) — Terminal T (VENT mode motor drive signal) • Are voltages **approx. 0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and airflow mode actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to the Step 19.
14*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (SHORT TO GROUND BETWEEN CLIMATE CONTROL UNIT AND AIRFLOW MODE ACTUATOR) OR ELSEWHERE** • Switch the ignition off. • Verify that continuity at the following harness-side connector terminals between climate control unit and ground. — Terminal V (DEFROSTER mode motor drive signal) — Terminal T (VENT mode motor drive signal) • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and airflow mode actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to the Step 19.
		No	Go to the next step.
15	**INSPECT AIRFLOW MODE ACTUATOR** • Inspect airflow mode actuator. (See AIRFLOW MODE ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is airflow mode actuator normal?	Yes	Go to the next step.
		No	Replace the airflow mode actuator. (See AIRFLOW MODE ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) Then go to the Step 19.
16	**INSPECT AIRFLOW MODE LINK** • Inspect airflow mode links. (See AIRFLOW MODE MAIN LINK REMOVAL/INSTALLATION.) — Is grease on link? — Are links securely and properly installed? — Are links free of obstructions and hindrances? • Are the above items normal?	Yes	Go to the next step.
		No	Apply grease to links. If any links are damaged, replace malfunctioning part. Then go to the Step 19.
17	**INSPECT TO SEE WHETHER MALFUNCTION IS IN CLIMATE CONTROL UNIT OR AIRFLOW MODE DOOR** • Inspect A/C unit airflow mode door. (See A/C UNIT REMOVAL/INSTALLATION.) — Is door free of obstructions, cracks and damage? — Are door securely and properly installed? • Are the above items normal?	Yes	Go to the next step.
		No	Remove the obstruction, or install door in proper position. If any doors are cracked or damaged, replace them. Then go to the Step 19. (See A/C UNIT DISASSEMBLY/ASSEMBLY.)
18	**VERIFY DUCT INSTALLATION CONDITION** • Are the following ducts installed properly? (See DASHBOARD REMOVAL.) (See DASHBOARD INSTALLATION.) — Dashboard ducts — Defroster ducts	Yes	Inspect ducts for restriction, deformity, and air leakage, then go to the next step.
		No	Install the incorrectly installed ducts properly, then go to the next step. (See DASHBOARD DISASSEMBLY/ASSEMBLY.)
19	**VERIFY THAT MALFUNCTION SYMPTOM DOES NOT RECUR AFTER REPAIR** • Is air discharged from vent?	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step 1 if the malfunction recurs.